



INTERNATIONAL ROADMAP FOR DEVICES AND SYSTEMS™

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2024 IRDS

PROLOGUE

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PROLOGUE

From the Chairman:

What a difference two years have made! We left 2022 with the recovery of the GDP in progress and with the electronics industry and the semiconductor industry rebalancing the supply chain to make up for what had been lost in 2020 due to the pandemic; it seemed as if the world was reconnecting with the 2019 environment and moving forward as if nothing had happened. However, all the sudden an architectural and technological revolution of unprecedented dimensions was announced to the crowds in November 2022 when Chat GPT was first released online: *Artificial Intelligence had finally arrived!*

After decades of failures, the long-awaited marriage of 100% parallel architecture with graphic processing units (GPU) devices was publicly celebrated. Not too many people knew at the time that this marriage had been made legal in 2009 by a group of students at Stanford. When Chat GPT reached 100 million users in two months it got the attention of the world on this new chapter of the electronics industry!

However, what was not clear to the electronics community was that the rules of the game had forever changed. “Over shrunk” channel length to make power hungry transistors switch faster was no longer desirable and instead, minimizing switch energy was the new rule—die size had reached the limits of $\sim 800 \text{ mm}^2$ imposed by lithography with Nvidia’s Hopper GPU microarchitecture and so, how was it possible to satisfy the insatiable hunger of AI for transistors now that these limits had been reached?

Salvation has come from a 60-year-old technology originally named “multi-chip module” (MCM) that has now become known by many other names in the industry, but it remains essentially the same. If it were not possible to place enough transistors in a single die to make the more than 208 billion transistors working together as required by the Blackwell GPU, it was however possible to achieve this by partitioning them in two dice co-located in a single package, thereby making all the transistors operate as if they were in a single die.

For the above reasons, the 2024 IRDS will include in addition to the Executive Summary a three-part Executive Tutorial on Packaging (EPT) followed later with a Tutorial on Data Centers (TDC).

Stay tuned!

Paolo Gargini